

Thesis Improvement Plan

Habituation-Aware Multi-Robot Intervention

1 Research direction and priorities

The submitted ACC paper establishes a useful starting point: intervention effectiveness depends on past actions, and combining effectiveness-aware valuation with flexible cue selection can reduce exposure under spare capacity. Its increased reactive misses and high-demand reversal motivate a thesis about allocating limited service capacity while preserving useful future intervention capability.

Recommended research question. When does a history-aware intervention deliver enough future exposure reduction to justify the robot time and reactive service it displaces?

The proposed extension has three core components: a consistent intervention forecast, action valuation through alternative service continuations, and admission that protects pending reactive deadlines. The purpose is to test whether these changes improve the exposure versus service trade-off. Better results are a hypothesis to evaluate, not a conclusion established by this memo.

What the submitted version motivates

Current formulation	Thesis development opportunity
Planning-time cue effectiveness; simplified historical suppression	Propagate recovery to application and reconstruct suppression from estimated application effectiveness.
Always-below-threshold exposure requirement	Check whether the common, unchangeable present dominates robustness; align the objective with accumulated exposure.
One-intervention counterfactual	Include the subsequent work delayed or displaced by that intervention.
Predictive reservation and urgency overrides	Admit predictive work according to benefit and pending-service feasibility.
Effectiveness floor	Evaluate preservation through the value of future cue use; retain a floor only when operationally justified.

Scope and evidence

This plan is based on the submitted manuscript to ACC 2027. That version specifies four UGVs, a 60×48 grid, and 1800-second runs with no warm-up. It also explicitly holds effectiveness constant during candidate travel. These details supersede earlier draft descriptions for this plan. The simulator code and raw results have not been audited here.

First milestone: establish a reproducible baseline and test whether forecast corrections and robustness masking materially change candidate rankings. Complete this before adding uncertainty estimation, reinforcement learning, or a large joint fleet optimizer.

2 Make the forecast consistent with the intervention dynamics

Let t be the decision time, $t_a = t + \delta_a$ the modeled application event, z_a the target cell, and c_a the cue. Define the application event consistently with the simulator: if suppression is inserted at service completion, δ_a includes the remaining travel and application hold. Distinguish arrival, application, completion, and replanning times explicitly.

Recovery before use and depletion after use

If no intervening application affects this cell–cue pair, use

$$\hat{q}_a = 1 - [1 - \hat{\eta}_{z_a, c_a}(t)] e^{-\delta_a / \hat{T}_{\text{rec}}}, \quad (1)$$

$$\hat{\eta}_{z_a, c_a}(t_a^+) = (1 - \hat{\kappa}) \hat{q}_a. \quad (2)$$

Known intervening applications must be processed chronologically, alternating recovery propagation with depletion jumps. Continue recovery after the candidate application. Use the same state-update routine for the internal planner state and its copied forecasts, with planner parameters kept separate from environment parameters.

Ignoring travel recovery understates application effectiveness under the assumptions of (1), but it does not establish conservative action selection. Its magnitude is

$$\hat{q}_a - \hat{\eta}_{z_a, c_a}(t) = [1 - \hat{\eta}_{z_a, c_a}(t)] (1 - e^{-\delta_a / \hat{T}_{\text{rec}}}).$$

Measure this difference across candidates before describing the approximation as small.

Historical suppression and shared forecast errors

Reconstruct existing suppression using the estimated effectiveness at each past application:

$$\hat{u}_r(x, t + s) = \sum_{j \in \mathcal{H}_r(t)} \hat{q}_j \beta_{c_j} K(x - x_j) e^{-(t+s-\tau_j)/\omega_u}, \quad s \geq 0.$$

Here $\mathcal{H}_r(t)$ contains only history available to robot r . The historical $\hat{q}_j$ is the estimate at application time, not today’s recovered effectiveness. Preserve the relevant excitation structure where available, and document any remaining approximation.

The submitted paper argues that shared baseline errors affect the absolute forecast rather than the counterfactual difference. This does not generally hold through clipping or nonlinear robustness. For example, with candidate suppression $h = 0.5$, a baseline intensity of 0.2 gives reduction $\min(0.2, 0.5) = 0.2$, whereas a baseline of 1 gives reduction 0.5. A common baseline error therefore changes action value.

Implementation checks. Use deterministic one-cell examples to verify recovery, pre-depletion suppression, post-application recovery, overlapping spatial effects, chronological neighbor updates, and equality of the two forecasts before the candidate application. Verify that forecasting never mutates the live state. Compare candidate rankings and predicted versus realized benefits before running a broad parameter sweep.

3 Align the objective and temporal requirements

Use accumulated exposure as the primary cost

For a continuation policy π and the robot-accessible information $\mathcal{I}_t$, define

$$\hat{J}_H(\pi \mid \mathcal{I}_t) = \hat{\mathbb{E}} \left[\int_t^{t+H} \sum_{z \in \mathcal{Z}} w_z \lambda_z^\pi(s) ds \mid \mathcal{I}_t \right]. \quad (3)$$

Here λ_z is a cell event rate, and w_z is its importance weight. Keep this convention distinct from a spatial density, which requires cell-area factors. A deterministic forecast is a first implementation; scenario averages can later approximate the expectation. A local implementation must specify its modeled region and how neighboring interventions affect that region. Local cost improvement does not imply fleet-wide improvement.

Diagnose the current robustness objective

For a hard temporal minimum, the robustness of $\Box_{[0,H]}(e_z \leq E^*)$ is determined by the largest exposure over the horizon. If the maximum is at the current time and both candidates only reduce future exposure, the clause assigns the same robustness to both. Smoothing the outer conjunction does not remove a hard minimum inside the clause.

Log which time, cell, and clause determine each candidate's score. Test a trace with identical current exposure and different future exposure. Also test whether changing inactive-action entries changes the number of soft-min arguments and thereby changes the score. Specify the mask, its numerical convention, and the aggregation tree explicitly; do not silently alter semantics to remove an inconvenient result.

Keep STL for operational temporal requirements

An alternative to always staying below a threshold is a timed recovery requirement:

$$\Box \left[(e_z > E_{\text{high}}) \Rightarrow \Diamond_{[0, T_{\text{response}}]} (e_z \leq E_{\text{low}}) \right], \quad E_{\text{low}} < E_{\text{high}}. \quad (4)$$

This is a proposed requirement, not an established property of the current method. A practical monitor must define trigger and reset rules, retain outstanding obligations across replans, and handle obligations extending beyond the current horizon. Preserve their original deadlines. Report infeasibility or a specified relaxation when the fleet cannot satisfy them. Restrict coverage requirements to operationally meaningful regions and revisit times. STL-constrained predictive control provides relevant background [2].

Value future cue use

A low-effectiveness cue can still be worth using during a severe event. A fresh cue can be worth preserving when a more valuable intervention is expected shortly. Carry depletion and recovery through subsequent forecast actions so these consequences enter (3). A horizon containing no later use cannot demonstrate the value of saving a cue for that later use. Compare against the original effectiveness-floor score; retain a hard floor only if the application truly requires one.

4 Compare service continuations and protect pending work

Let π_0 be a feasible baseline continuation, initially the habituation-aware greedy policy B5. Construct π_a by applying candidate a and then resuming the same baseline decision rule from the resulting state. The continuation must be recomputed, because the candidate changes robot availability, task queues, bird intensity, and cue effectiveness.

$$U(a) = \hat{J}_H(\pi_0) - \hat{J}_H(\pi_a). \quad (5)$$

Both continuations use the same horizon, initial information, and assumptions about neighboring commitments. In stochastic rollouts, share exogenous random streams while allowing action-dependent intensities and detections to evolve differently. Identical realized task lists would suppress part of the intervention’s mission effect.

A manageable first implementation

Enumerate a limited set of target–cue choices and simulate each followed by B5. Include the unchanged baseline. Model travel, service holds, queued responses, and at least some subsequent interventions within the horizon. Use a fixed time horizon rather than different elapsed horizons for different candidates. Record the treatment of unfinished tasks and terminal state. Limited rollout has a relevant multiagent precedent [3]; its formal improvement guarantees do not transfer automatically to an approximate decentralized simulator.

Service-aware candidate selection

Let $\mathcal{Q}_t$ be the pending reactive tasks, d_j their deadlines, and $\hat{C}_j(\pi)$ their forecast service completion times. First protect tasks that the baseline predicts it can complete on time:

$$\mathcal{F}_0 = \{j \in \mathcal{Q}_t : \hat{C}_j(\pi_0) \leq d_j\}.$$

Use the finite candidate problem

$$\begin{aligned} \pi^* &\in \arg \min_{\pi \in \Pi_{\text{cand}}} \hat{J}_H(\pi) \\ \text{subject to } &\hat{C}_j(\pi) \leq d_j, & j \in \mathcal{F}_0, \\ &\text{motion, service, and cue dynamics,} \end{aligned} \quad (6)$$

with $\pi_0 \in \Pi_{\text{cand}}$. For baseline-late tasks with finite completion times, an optional additional condition is $\hat{C}_j(\pi) \leq \hat{C}_j(\pi_0)$. A dropped or uncompleted task is not a successful service; define its completion as infinite and report it. Forecast far enough to assess each protected deadline, or retain a verified completion schedule beyond the exposure horizon.

Admit a predictive deviation only when it is feasible and $U(a) > \varepsilon_{\text{gate}} \geq 0$; otherwise execute the baseline action. Treat predictive reservation as a ceiling, or remove it, rather than forcing a minimum predictive workload. The gate should reflect prediction accuracy and a specified operational preference, not post hoc tuning on test seeds.

These constraints protect predicted service for known tasks. They do not guarantee deadlines under unexpected arrivals or travel error. Evaluate such disturbances separately. Concurrent robots must exchange or reserve commitments; individually acceptable changes can conflict when executed together.

5 Algorithm and achievable analytical claims

Decision procedure

1. Reconstruct the current local state from available observations and intervention history. Freeze one snapshot for the decision.
2. Generate the baseline continuation, its exposure cost, and its pending-task completion predictions.
3. Generate target-cue candidates. Forecast each candidate followed by the baseline rule using copied states and consistent scenario streams.
4. Reject candidates violating motion, service, or protected-deadline conditions. Apply any chosen obligation-relaxation policy explicitly.
5. Select the lowest-cost remaining continuation. Execute its first action only if its improvement exceeds the admission margin; otherwise use the baseline.
6. Record predicted benefit, active constraints, realized service, and runtime. Replan from new observations while preserving existing temporal obligations.

Proposition 1 (Predicted baseline comparison). *At a fixed decision state, suppose the unchanged baseline is in the feasible candidate set and all candidates are evaluated with the same cost definition and horizon. An exact minimizer of (6) satisfies*

$$\hat{J}_H(\pi^*) \leq \hat{J}_H(\pi_0).$$

Proof. The baseline is an available feasible choice, so the minimum cannot exceed its evaluated cost. A positive admission margin preserves this inequality because unsuccessful deviations revert to the baseline. $\square$

This is an elementary design property, not a standalone novelty claim. It is useful because the original dispatcher can execute a predictive candidate with negative valuation. Establishing a closed-loop result under repeated replanning, incomplete information, and concurrent decisions requires additional assumptions and analysis.

Proposition 2 (Effect of bounded cost-estimation error). *At a fixed decision state, suppose $|\hat{J}_H(\pi) - J_H(\pi)| \leq \epsilon$ for every evaluated candidate and the baseline, where J_H is the true expected cost of executing that entire continuation over the same horizon. Then a selected candidate with no greater predicted cost satisfies*

$$J_H(\pi^*) \leq J_H(\pi_0) + 2\epsilon.$$

If $\hat{J}_H(\pi_0) - \hat{J}_H(\pi^) > 2\epsilon$, its true continuation cost is strictly smaller.*

Proof. $J_H(\pi^*) \leq \hat{J}_H(\pi^*) + \epsilon$ and $\hat{J}_H(\pi_0) \leq J_H(\pi_0) + \epsilon$. Combining these inequalities yields both statements. $\square$

The substantive research task is to justify or empirically characterize estimation error, not to assume a small ϵ . A validation-set error quantile is not a deterministic uniform bound. A guarantee about expected continuation cost is also not a guarantee for each random realization or the eventual replanned trajectory.

6 Experiments that identify the source of improvement

Use successive controlled comparisons. Start with spare and high demand at the confirmatory depletion level, then broaden only after the diagnostic questions are resolved. Keep candidate generation, information access, and cue availability matched wherever the comparison requires it.

Variant	Definition	Principal comparison
A	Submitted B4 implementation	Reproduction reference.
B	A with a consistent forecast	B versus A isolates the forecast changes.
C	B with integrated-exposure valuation and the same dispatch/cues	C versus B tests objective alignment.
D	Baseline-continuation exposure rollout, with positive-benefit admission	D versus C tests the combined rollout/admission redesign.
E	D with pending-service protection	E versus D isolates the service constraint.
B5	Habituation-aware greedy baseline	E versus B5 tests practical system benefit.

D versus C changes continuation modeling, admission, and predictive scheduling. If this contrast is important, add C with the same positive-benefit gate to separate those contributions. If retaining STL response constraints in E, add an otherwise identical version without those constraints. Do not attribute a system-level improvement to one component when several changed.

Outcomes and diagnostics

- **Mission:** integrated weighted exposure, its paired difference from B5, and the exposure versus service trade-off.
- **Service:** miss fraction, absolute missed-task count, task generation, completion, and time overdue. Define response time for completed tasks and report unfinished tasks separately to avoid survivorship bias.
- **Mechanism:** candidate ranking changes, prediction error, rejected predictive actions, travel time, effectiveness at application, and future cue reuse.
- **Implementation:** decision runtime and its upper quantiles, candidate count, communication assumptions, and behavior when computation times out.

Persistent operation and fair comparisons

The submitted runs last 1800 seconds without warm-up. Extend evaluation to several actual recovery time constants, with initially fresh and initially depleted cues, and with demand changes within a run. Report startup and later windows separately; do not assume stationarity merely from a long run. Use the recovered production value of T_{rec} , not an inferred default.

Separate development seeds from final evaluation seeds. Couple exogenous randomness consistently across policies, preserve complete seed pairs in bootstrap intervals, and state the percentage-effect formula and multiplicity family. To claim service noninferiority, specify an operationally meaningful margin in advance and show the appropriate confidence bound; a nonsignificant difference is insufficient. Test zero habituation, overload, parameter mismatch, and delayed or missing neighbor histories. A fixed load-switching threshold near the paper’s observed reversal is not yet justified.

7 Work packages, thesis structure, and claim boundaries

1. **Reproduce and diagnose.** Archive the submitted implementation and exact configurations, including monitor settings, cue count/strengths, units, deadline rules, and timing order. Deliver a reproduction table and candidate-level traces that reveal forecast error and robustness masking.
2. **Correct the forecast.** Implement shared state-propagation routines and chronological action histories. Deliver deterministic checks and the A-versus-B comparison, including cases where rankings do not change.
3. **Implement the proposed planner.** Build exposure-based continuations, positive-benefit admission, and protection for pending service. Deliver a clear algorithm, runtime measurements, and documented fallback behavior.
4. **Evaluate and analyze.** Run the controlled progression, assess prolonged operation, and state which benefits survive uncertainty and higher load. Develop the conditional bounds with explicit assumptions.

Organize the thesis around the existing method, forecast and valuation diagnostics, proposed planner, analytical properties, and controlled evaluation. The conference paper motivates the new design.

Claims to target

The central empirical target is: *The planner exploits habituation to reduce exposure while limiting the additional reactive-service disruption caused by predictive interventions, under specified demand and modeling conditions.* Establish each part from paired comparisons and a predeclared service criterion.

Keep theoretical and empirical claims distinct. Predicted baseline improvement follows from the candidate-set construction. True improvement requires justified error assumptions or empirical evidence. Global deadline guarantees, universal superiority to greedy dispatch, species-specific deterrence effectiveness, and decentralized scalability require additional evidence beyond the proposed first implementation.

Report which component produces the gain. If exposure-based scoring matches STL, describe STL as a requirement language. If service protection removes the exposure advantage, characterize that capacity limitation and the achievable trade-off.

Recommended minimum scope: consistent forecasting, continuation-based exposure valuation, service-aware admission, and a controlled evaluation with conditional analytical statements. Online parameter learning and field calibration are later extensions unless the core results show that they are necessary.

References

- [1] G. Basus and S. Farzan, “Habituation-aware task valuation for multi-robot bird deterrence using signal temporal logic,” manuscript submitted to the 2027 American Control Conference, 2026.
- [2] V. Raman, A. Donzé, M. Maasoumy, R. M. Murray, A. Sangiovanni-Vincentelli, and S. A. Seshia, “Model predictive control with signal temporal logic specifications,” in *Proc. IEEE Conference on Decision and Control*, 2014, pp. 81–87. <https://people.eecs.berkeley.edu/~sseshia/pubs/b2hd-raman-cdc14.html>.
- [3] D. Bertsekas, “Multiagent rollout algorithms and reinforcement learning,” arXiv:1910.00120, version 3, 2020. <https://arxiv.org/abs/1910.00120>.